

Topic 1 Review [100 marks]

1. [Maximum mark: 4] 25M.1.SL.TZ2.3
Describe **two** advantages of using surveys to determine user requirements for updating a computer system. [4]

2. [Maximum mark: 2] 25M.1.SL.TZ2.6
Identify **two** methods of implementing a new computer system. [2]

3. [Maximum mark: 2] 25M.1.SL.TZ2.7
Describe the role of a DNS server. [2]

4. [Maximum mark: 15] 25M.1.SL.TZ2.9
An architect uses a single computer for her business. She uses hand-drawn sketches to present her ideas to clients and computer-aided design (CAD) software to create construction projects, alterations, and redevelopments.

(a.i) State what is meant by CAD software. [1]

(a.ii) Outline **two** benefits for the architect of using CAD software. [4]

The architect often creates large graphic files that are sent via the internet.

(b) Explain the need for data compression software in storing and transferring these large graphic files. [3]

Data loss can cause downtime, which can force the architect to pause business operations.

The impact on productivity depends on the amount of data lost as well as the time it takes for data recovery.

(c.i) Describe how the architect could minimize downtime if the disk system on her computer fails. [3]

(c.ii) Outline **two** causes of data loss **other than** hardware failure. [4]

5. [Maximum mark: 15] 19N.1.SL.TZ0.8

An organization is implementing a new computer system.

(a) Identify **two** organizational issues related to the implementation of the new system. [2]

The management considered phased conversion and direct changeover as methods of implementation.

(b) Evaluate these **two** methods of implementation. [5]

(c.i) State **one** type of testing that involves users. [1]

(c.ii) Identify **three** consequences of inadequate testing. [3]

(d) Discuss the social and ethical issues associated with the introduction of a new computer system. [4]

6. [Maximum mark: 3] 19N.1.SL.TZ0.2

(a) Identify **one** cause of data loss. [1]

(b) Describe **one** way offsite storage can be used to prevent data loss. [2]

7. [Maximum mark: 15]

20N.1.SL.TZ0.8

Website developers need to consider a range of usability factors when designing a website.

(a) Identify **two** usability factors that need to be considered in the design of a website. [2]

(b) Outline **one** reason why visual displays on a computer screen can create difficulties for some people. [2]

A company promotes its products online. To make a purchase, customers are required to register with the company and provide data like their name, date of birth, age, gender and email address. Once registered, more than one customer is able to access the server to retrieve and modify their data at the same time.

(c.i) State where the customer data is held during the process of modifying their data. [1]

(c.ii) Explain how the operating system ensures that each customer's data is secure when multiple users are accessing the data at the same time. [4]

(d) The company is considering sharing its customers' data with marketing organizations.

Explain why there could be ethical issues for the company when sharing its customers' data. [6]

8. [Maximum mark: 15]

20N.1.SL.TZ0.7

A school has a local area network (LAN) connecting its computers and peripheral devices. The LAN also provides access to the internet.

(a) Describe the role of a router in this network. [3]

Users have been troubled by slow speeds when accessing the internet.

(b) Outline **two** reasons why there might be a reduction in data transmission speed at certain times. [4]

(c) Outline **two** measures that the school could take to safeguard its data from unlawful access via the internet. [4]

(d) The inventory of office supplies used in the school is stored on the computer as a single file.

Each of the office supplies in the inventory (such as paper, ink, toner, printers, pens, staplers, pencils and scissors) has a unique ID number, name, maximum quantity, minimum quantity and remaining quantity.

Outline the steps in an algorithm that would output a list of supplies with the quantity to be ordered. [4]

9. [Maximum mark: 6] 20N.1.SL.TZ0.2

An airline has a server that holds the flight database. Passengers can check in using a number of self-service client kiosks located in the airport.

(a.i) Define the term *client*. [1]

(a.ii) Define the term *server*. [1]

(b) Explain the functions performed by the server in this situation. [4]

10. [Maximum mark: 3] 20N.1.SL.TZ0.1

(a) Outline **one** feature of a word processor that could reduce the amount of typing required when writing letters. [2]

(b) State the purpose of technical documentation provided with software. [1]

- 11.** [Maximum mark: 1] 21M.1.SL.TZ0.10
Identify **one** common feature found in the user interface of application software to improve its usability. [1]
- 12.** [Maximum mark: 4] 21M.1.SL.TZ0.4
(a) Identify **two** reasons why patches may be necessary for an operating system. [2]
(b) Identify **two** methods that can be used to obtain these patches. [2]
- 13.** [Maximum mark: 2] 21M.1.SL.TZ0.1
Identify **two** roles that a computer can perform in a network. [2]
- 14.** [Maximum mark: 2] 21M.1.SL.TZ0.2
Describe **one** method of implementation for a new computer system. [2]
- 15.** [Maximum mark: 7] 21N.1.HL.TZ0.7
A group of programmers are involved in creating a new software product. They create many new sub-programs but also use existing sub-programs within the product.
- (a) Outline why a sub-program is considered an example of abstraction. [2]
- (b) Evaluate the use of designing and developing different parts of software products concurrently. [3]

- (c) Outline **one** way in which users can be informed of software updates. [2]

16. [Maximum mark: 2] 22M.1.SL.TZ0.1
State **two** compatibility issues that may arise when international businesses merge. [2]

17. [Maximum mark: 2] 22M.1.SL.TZ0.2
Outline **one** reason why accurate user documentation is necessary for a system. [2]